

Online Retail Sales & Customer Analytics

Data Cleaning • Exploratory Analysis • SQL • RFM Customer Segmentation

Portfolio Project | 2026

£10.6M	19,960	4,338	524,878
Cleaned sales revenue	Distinct orders	Identified customers	Cleaned transaction lines

Purpose: demonstrate an end-to-end customer analytics workflow using a retail transaction dataset, from data preparation through customer segmentation and visualization.

1. Project Objective & Analytical Workflow

Objective

Analyze online retail transaction data to understand sales patterns, geographic concentration, product performance, customer purchasing behavior, and customer value using RFM segmentation.

Workflow

1	Data preparation	Remove exact duplicates, cancelled invoices, and non-positive quantity/price records.
2	Derived measures	Calculate Revenue = Quantity × UnitPrice and retain missing CustomerID rows for overall sales analysis.
3	EDA & SQL	Analyze revenue, orders, products, countries, customers, and purchasing frequency.
4	Customer analysis	Use identified customers for customer-level revenue and order-frequency analysis.
5	RFM segmentation	Calculate Recency, Frequency, and Monetary measures and apply the supplied segment rules.
6	Visualization	Present descriptive findings in a portfolio dashboard/report.

Analytical boundary: overall sales includes transactions without CustomerID; customer-level and RFM analysis uses identified customers only.

2. Data Preparation & Scope

Final cleaned dataset

Measure	Value
Raw transaction rows	541,909
Exact duplicate rows removed	5,268
Final cleaned sales rows	524,878
Cleaned sales revenue	£10,642,110.80
Distinct orders	19,960
Identified customers	4,338
Products / stock codes	3,922

Cleaning rules

- Exact duplicate rows removed.
- Cancelled invoices removed where InvoiceNo begins with C.
- Rows with Quantity ≤ 0 removed.
- Rows with UnitPrice ≤ 0 removed.
- Revenue derived as Quantity $\times$ UnitPrice.
- Missing CustomerID rows retained for overall sales analysis but excluded from customer-level and RFM analysis.
- Service/adjustment lines are retained in overall sales; merchandise-only product rankings use the defined exclusions.

Scope note: the first and last calendar years are partial, so year-to-year totals should not be interpreted as full-year comparisons.

3. Sales Overview

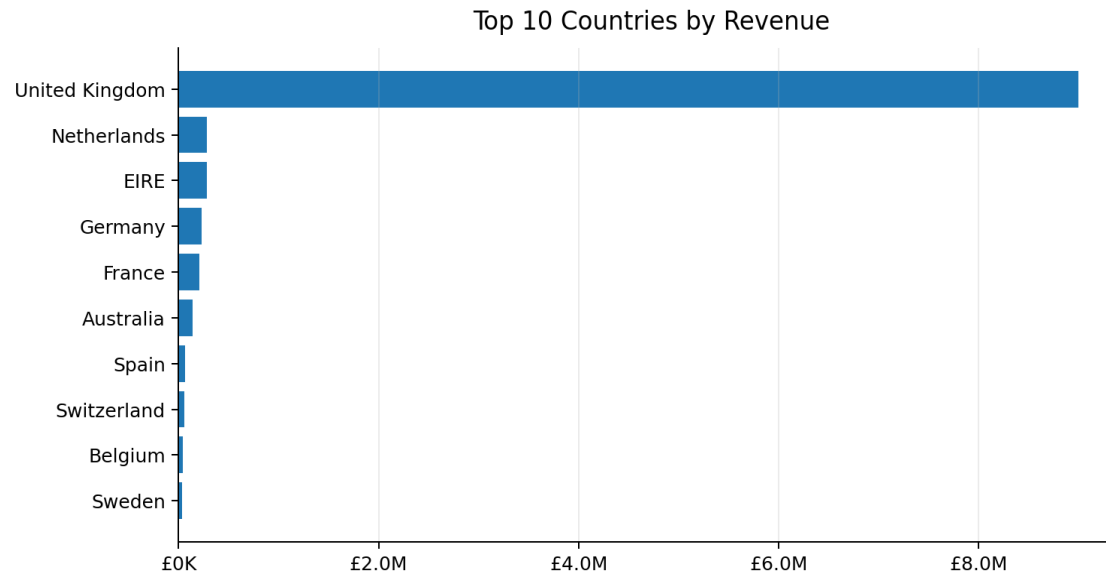
£10.6M	19,960	5,572,420	£533.17
Total revenue	Distinct orders	Units sold	Average order value

The cleaned transaction set contains 524,878 sales lines and £10.64M in revenue. Monthly and annual views are descriptive; boundary years are incomplete.

Interpretation

The dataset shows substantial variation in monthly sales volume. Because the available period begins and ends partway through calendar years, the report avoids treating 2010 and 2011 as directly comparable full-year periods.

4. Geographic Sales Analysis

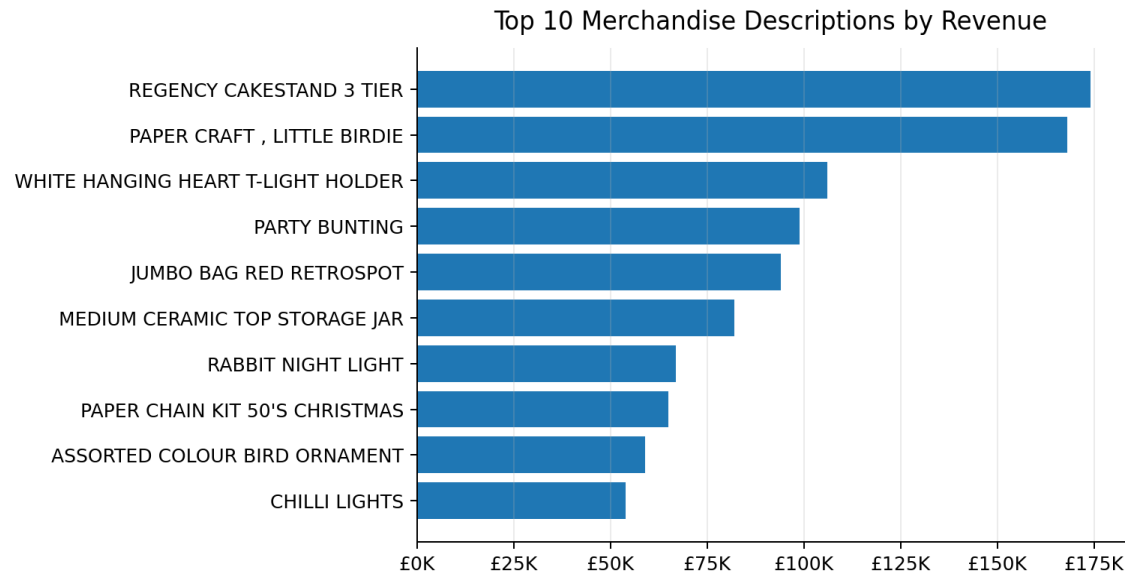


Country	Revenue	Share
United Kingdom	£9,001,744	84.6%
Netherlands	£285,446	2.7%
EIRE	£283,141	2.7%
Germany	£228,678	2.1%
France	£209,625	2.0%
Australia	£138,454	1.3%
Spain	£61,559	0.6%
Switzerland	£57,068	0.5%
Belgium	£41,196	0.4%
Sweden	£38,368	0.4%

The United Kingdom accounts for 84.6% of cleaned sales revenue. This is a descriptive concentration measure, not a causal explanation of market performance.

5. Product & Merchandise Analysis

Merchandise revenue ranking



The merchandise ranking excludes selected service and adjustment codes. Under the all-lines definition, DOTCOM POSTAGE is the highest-revenue description; therefore the merchandise leader should not be described as the overall revenue leader.

Top merchandise quantities

Description	Units
PAPER CRAFT , LITTLE BIRDIE	80,995
MEDIUM CERAMIC TOP STORAGE JAR	78,033
WORLD WAR 2 GLIDERS ASSTD DESIGNS	54,951
JUMBO BAG RED RETROSPOT	48,371
WHITE HANGING HEART T-LIGHT HOLDER	37,872
POPCORN HOLDER	36,749
PACK OF 72 RETROSPOT CAKE CASES	36,396

Description	Units
ASSORTED COLOUR BIRD ORNAMENT	36,362
RABBIT NIGHT LIGHT	30,739
MINI PAINT SET VINTAGE	26,633

6. Customer Purchasing Behavior

4,338	2,845	1,493	65.6%
Identified customers	Repeat customers	One-time customers	Repeat-customer share

Customer-level revenue and order-frequency analysis uses identified customers only. One-time customers account for 34.4% of identified customers, while repeat customers account for 65.6%.

Important scope distinction

Customer revenue is not the same analytical measure as total cleaned sales revenue because unidentified transactions remain in the overall sales dataset.

7. RFM Customer Segmentation

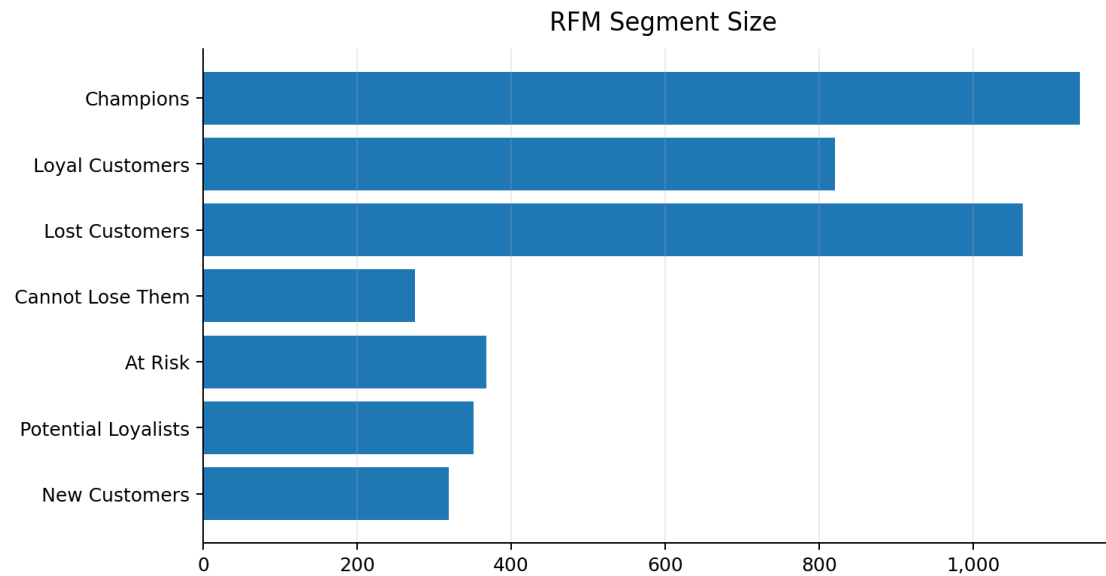
RFM framework

Measure	Definition
Recency	Days since the customer's most recent purchase, using the analysis date defined in the notebook.
Frequency	Number of distinct invoices/orders associated with the customer.
Monetary	Total customer revenue.
Scoring	Customers are assigned R, F and M scores on a 1–5 scale using the project's quantile-based scoring approach.
Segments	Seven supplied customer labels: Champions, Loyal Customers, Lost Customers, Cannot Lose Them, At Risk, Potential Loyalists, New Customers.

RFM summary

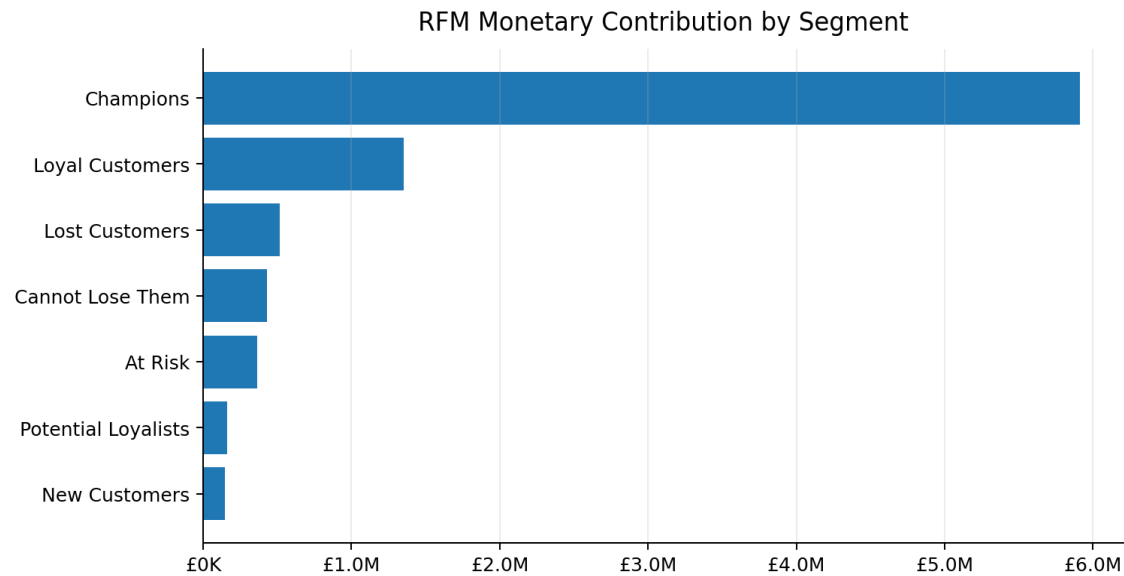
Total RFM customers	4,338
Total RFM monetary value	£8.89M
Average recency	92.54 days
Average frequency	4.27
Average monetary value	£2,048.69
Average RFM total	9.02

8. RFM Segment Distribution



Segment	Customers	Share of RFM customers	RFM monetary	Monetary share
Champions	1,139	26.3%	£5,913,422	66.5%
Loyal Customers	821	18.9%	£1,351,084	15.2%
Lost Customers	1,065	24.6%	£518,322	5.8%
Cannot Lose Them	275	6.3%	£431,639	4.9%
At Risk	368	8.5%	£366,413	4.1%
Potential Loyalists	351	8.1%	£161,108	1.8%
New Customers	319	7.4%	£145,220	1.6%

9. RFM Monetary Contribution



The supplied RFM file shows £8.89M in customer-level monetary value. Champions contribute 66.5% of that RFM monetary total, followed by Loyal Customers at 15.2%. These percentages describe the supplied customer-level RFM dataset and should not be compared directly with the £10.64M overall sales total.

The report intentionally avoids causal or forward-looking claims. Segment labels describe the customer-level scoring output rather than predicting future behavior.

10. RFM Segment Scorecard

Segment	Customers	RFM Monetary	Monetary %	Avg Recency	Avg Frequency	Avg Monetary	Avg RFM
Champions	1,139	£5,913,422	66.5%	13.31	9.99	£5,192	13.50
Loyal Customers	821	£1,351,084	15.2%	38.04	3.64	£1,646	10.39
Lost Customers	1,065	£518,322	5.8%	217.90	1.10	£486.69	4.71
Cannot Lose Them	275	£431,639	4.9%	137.08	4.89	£1,570	9.70
At Risk	368	£366,413	4.1%	164.63	2.29	£995.69	7.36
Potential Loyalists	351	£161,108	1.8%	53.50	1.16	£459.00	6.56
New Customers	319	£145,220	1.6%	18.52	1.24	£455.23	7.84
Total	4,338	£8,887,209					

Average columns are customer-level means and should not be summed across segments. The table is sorted by RFM monetary contribution, not by customer quality.

11. Key Findings & Limitations

Descriptive findings

- Cleaned sales revenue is £10.64M across 524,878 transaction lines and 19,960 distinct orders.
- The United Kingdom contributes 84.6% of cleaned sales revenue.
- REGENCY CAKESTAND 3 TIER is the highest-revenue merchandise description under the defined merchandise-only ranking.
- 4,338 customers are identified; 65.6% are repeat customers based on distinct-order behavior.
- RFM analysis contains 4,338 identified customers and £8.89M in customer-level monetary value.
- Champions contribute 66.5% of supplied RFM monetary value.

Limitations & analytical notes

- First and last calendar years are partial.
- Missing CustomerID transactions remain in overall sales but cannot be attributed to an identified customer for RFM.
- Merchandise rankings use defined service/adjustment-code exclusions; overall sales totals do not.
- Country names and stock codes are used as stored in the supplied data.
- The dataset does not contain a separate currency field; GBP formatting follows the project context.
- The analysis is descriptive and does not establish causality or forecast future customer behavior.

Tools used: Python / Pandas • SQL • Power BI / dashboard visualization

Source: Online Retail transaction dataset and project-generated cleaned/RFM outputs.